Global Weather Trend Forecasting

Advanced Data Science Assessment Report

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PM Accelerator • October 2026

This project uses global daily weather data to compare forecasting models, find unusual weather observations, and look at weather and air-quality patterns across different locations.

PM Accelerator mission

PM Accelerator mission is to make industry-leading tools and education available to individuals from all backgrounds, helping level the playing field for future PM leaders. It focuses on giving aspiring and experienced PMs access to industry leaders, the right PM ecosystem, and AI product management skills.

1. Project objective

The goal was to use the Global Weather Repository to forecast temperature and explore the data in more depth. I cleaned the data, compared several models, detected unusual observations, and looked at feature importance, air quality, and geographic patterns.

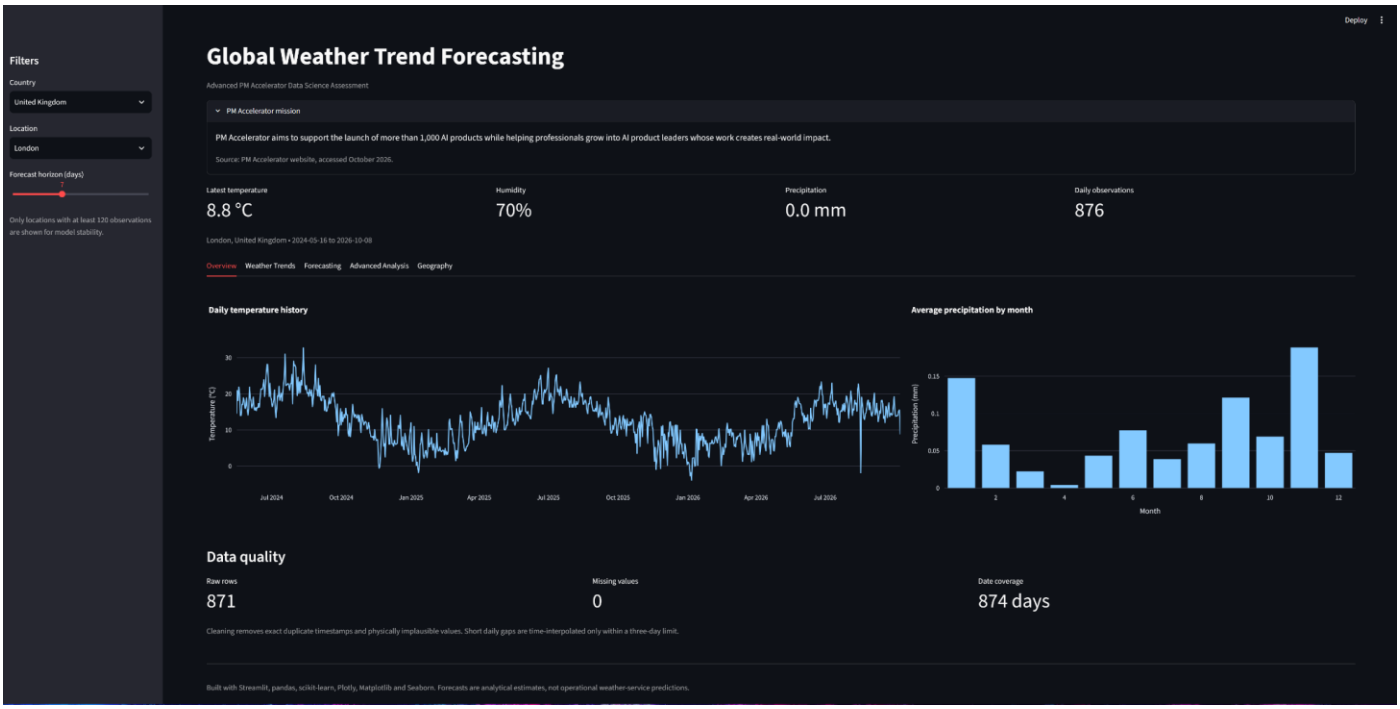


Figure 1. Main Streamlit dashboard for the selected city. The controls on the left let the user change the country, location, and forecast horizon.

2. Dataset and preprocessing

The dataset contains daily weather observations from cities around the world. It includes temperature, humidity, wind, pressure, precipitation, cloud cover, visibility, UV index, coordinates, and air-quality measurements. The data runs from May 2024 through October 2026. That is enough for this forecasting project, but not enough to make strong claims about long-term climate change.

I converted the timestamps, sorted each city by date, removed duplicate records, and checked for unrealistic values. I also filled only short gaps in the daily series. For forecasting, I used past temperature values, rolling averages, and date-based features. All of these were shifted so the model only used information that would have been available before the day being predicted.

3. Exploratory analysis

I first looked at temperature over time for each city. London shows a clear seasonal pattern, and the 30-day moving average makes it easier to see. The raw daily line also shows a few sharp changes that were worth checking later.

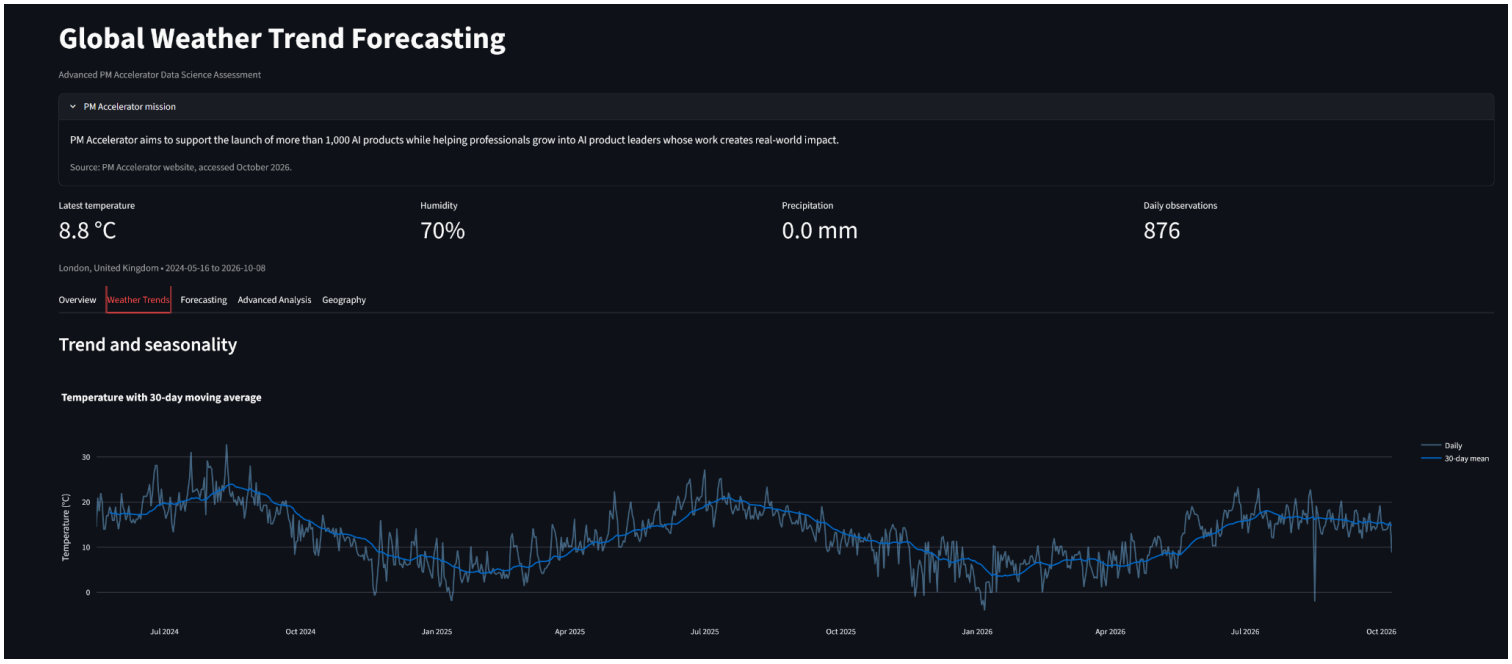


Figure 2. Daily London temperature with a 30-day moving average. The smoothed line makes the seasonal pattern easier to see.

The correlation matrix gives a quick view of how the weather and air-quality variables relate to each other. Temperature and UV index move together, while PM2.5 and PM10 have a strong positive relationship. I treated these as correlations only, not proof that one variable causes another.

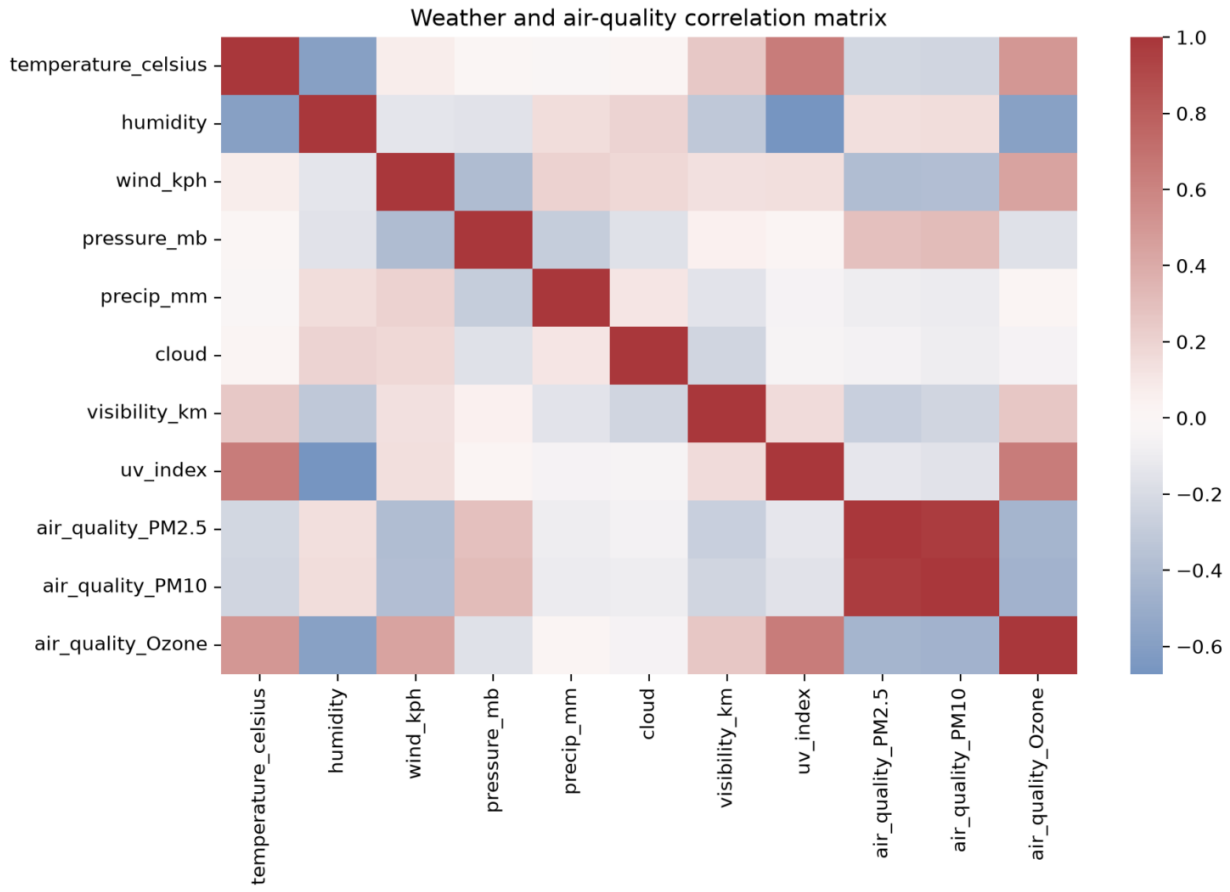


Figure 3. Correlation matrix for weather and air-quality variables.

4. Forecasting models and validation

I compared a seven-day seasonal naive baseline with Ridge Regression, Random Forest, Gradient Boosting, and an ensemble of the three trained models. I used the most recent 20% of the data as the test period instead of a random split. A random split would mix older and newer dates and could make the results look better than they really are.

For London, the ensemble performed best with an MAE of 1.95 °C and an RMSE of 2.79 °C. The other trained models were close, while the seasonal naive baseline was worse. I focused mainly on MAE and RMSE because MAPE can become misleading when the actual temperature is close to zero.



Figure 4. London holdout model comparison and actual-versus-predicted temperatures. The ensemble produced the best RMSE and MAE for this location.

The dashboard also shows a seven-day forecast from each trained model. The lines are not identical, but they stay in a similar range. This makes it easy to compare how each model expects the temperature to change over the next week.

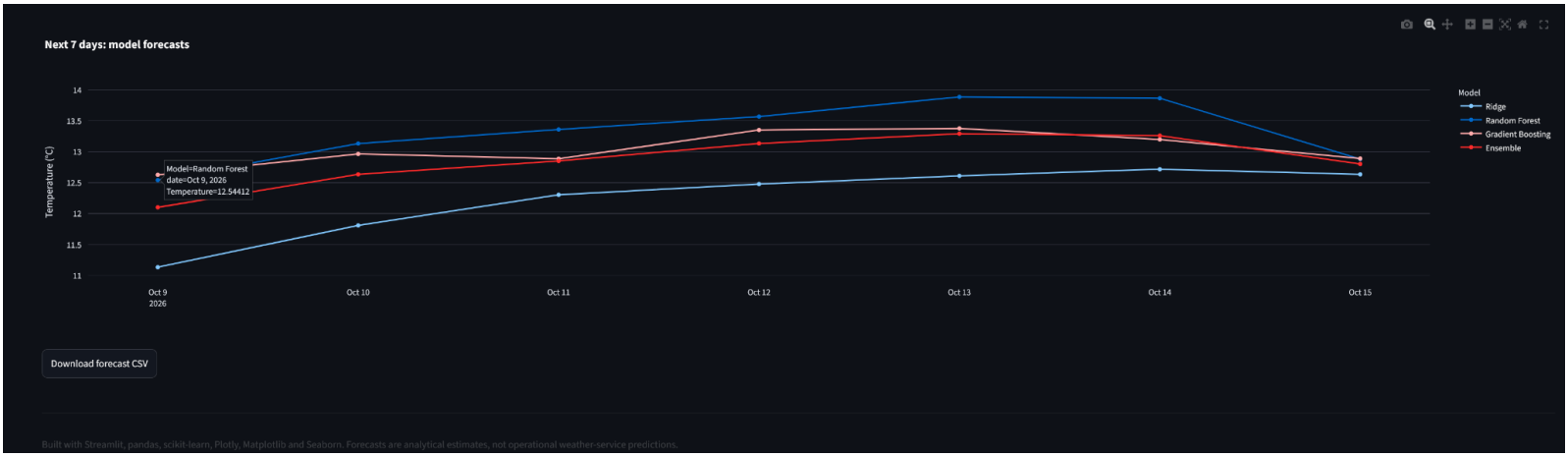


Figure 5. Seven-day temperature forecasts from Ridge, Random Forest, Gradient Boosting, and the ensemble.

5. Advanced analysis

Anomaly detection

I used Isolation Forest to flag unusual observations using several weather variables at once. For London, 27 observations were flagged at about 3% contamination. I did not automatically remove them because some could be real weather events while others could be bad data.

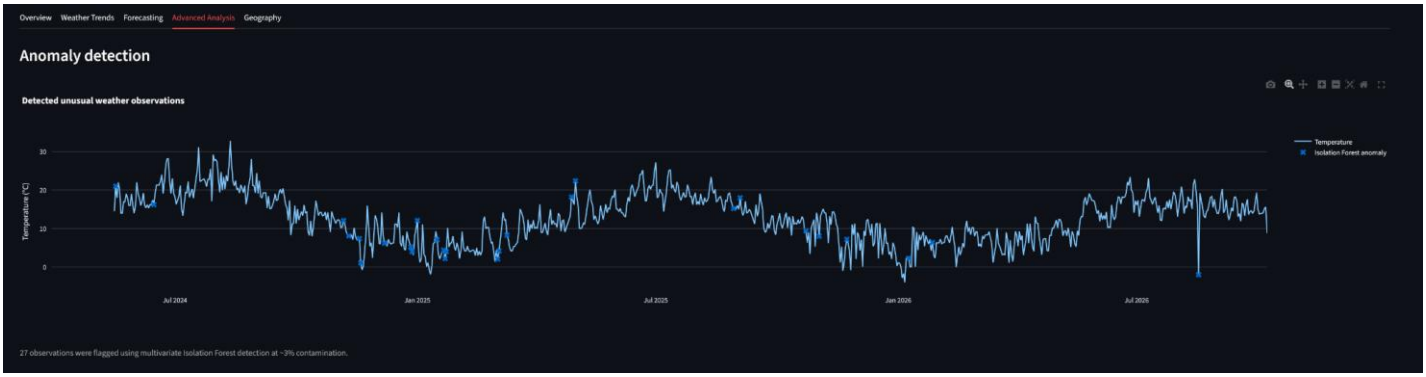


Figure 6. Isolation Forest anomaly detection for London. Blue markers identify observations that were flagged as unusual by the multivariable model.

Feature importance and environmental analysis

I used both Random Forest importance and permutation importance to compare which variables had the most influence. The two methods did not rank everything the same way, which is why I kept both. I also compared PM2.5 with other weather and pollution variables. In London, PM10 and nitrogen dioxide had the strongest positive relationships with PM2.5, while wind and ozone were more negative.



Figure 7. Two feature-importance methods and the local PM2.5 correlation analysis.

6. Geographic patterns

The geography section looks beyond one city. The map shows the latest available weather reading for each location, while the bar charts compare average temperature and PM2.5 by continent. In this dataset, Europe had the lowest average temperature, while Asia had the highest average PM2.5. These results only describe the dates included in the dataset.

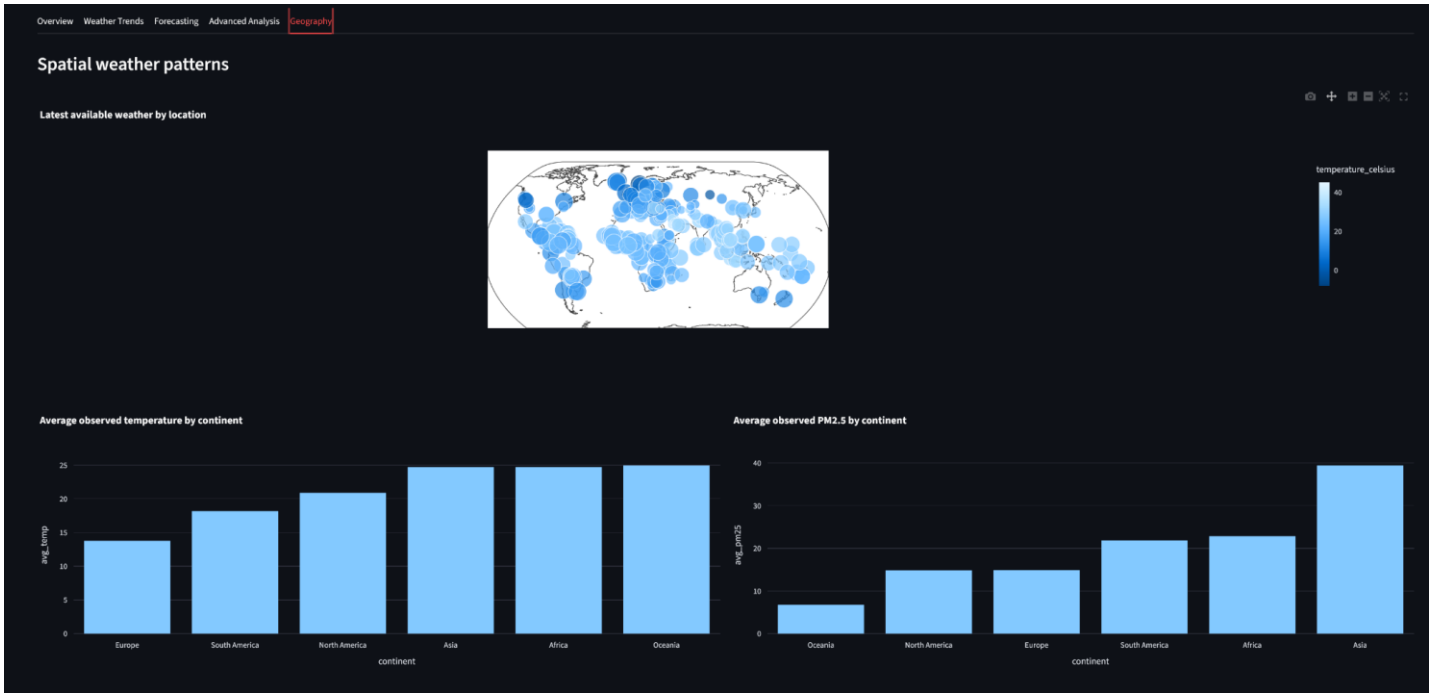


Figure 8. Spatial weather map plus continent-level comparisons for temperature and PM2.5.

7. Representative model results

I tested the forecasting pipeline on several cities with long and mostly complete histories. The best model changed by location. The ensemble worked best for London and Paris, while Ridge Regression worked better for Tokyo, Cairo, and Singapore.

Location	Best model	MAE (°C)	RMSE (°C)	MAPE
London, United Kingdom	Ensemble	1.95	2.79	20.8%
Tokyo, Japan	Ridge	2.64	3.44	10.9%
Cairo, Egypt	Ridge	1.70	2.45	6.9%
Paris, France	Ensemble	2.11	2.89	25.4%
Singapore, Singapore	Ridge	1.26	1.80	4.2%

MAPE was higher for London and Paris because percentage error can jump when the actual temperature is close to zero. I relied more on MAE and RMSE when comparing the models.

8. Conclusion

The project showed that the best forecasting model depends on the location. The ensemble worked best for London and Paris, while Ridge Regression did better in Tokyo, Cairo, and Singapore. The dashboard also made it easy to compare forecasts, spot unusual observations, and look at weather and air-quality patterns in one place. The main limitation is the short time span. A little over two years works for this assessment and short-term forecasting, but it is not enough for long-term climate conclusions. With more time, I would add older

weather data and test the models across several different time periods instead of using only one final test period.